Digital Campus Library Reservation Gateway

Requirements Table | Problem Statement #08

Req ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall allow a student member to search the ISBN catalog by title, author, or ISBN and view paginated availability results.	High	For a seeded catalog of 50,000 titles, each valid query returns matching records with title, ISBN, availability, and page controls; invalid/empty queries show a validation message.	Search is the primary entry point for locating library material.
FR-002	Functional	The system shall allow a student member to place a hold on a currently loaned book and assign the request the next FIFO queue position.	High	Given an unavailable title, the first valid request receives position 1 and later valid requests receive increasing positions; a member at the configured active-hold limit is rejected.	FIFO reservations provide fair access to scarce copies and enforce lending policy.
FR-003	Functional	The system shall notify the next eligible student member when a reserved copy becomes available and record the notification expiry time.	High	When a copy is returned, exactly the next eligible member receives an in-app and email notification within 60 seconds, with a configurable pickup deadline stored.	Timely notification converts queue movement into an actionable pickup.
FR-004	Functional	The system shall allow the head librarian to check out, return, and mark a reserved copy as collected or expired.	High	For each transaction, the system updates copy status, borrower/reservation state, and timestamp; a copy cannot be checked out twice concurrently.	Librarian operations maintain accurate inventory and reservation state.
FR-005	Functional	The system shall calculate, display, and record overdue fines for items returned after their due date.	Medium	For a returned item, fine = configured daily rate x overdue days; the recorded amount, member, item, and payment status are visible in the transaction history.	Fine records support transparent, auditable overdue handling.
NFR-001	Nonfunctional	The catalog search query shall return paginated results for 50,000 titles in less than 150 ms at the 95th percentile under simulated peak load.	High	A benchmark with 100 concurrent authenticated users records p95 response time below 150 ms for at least 1,000 representative queries, with no data-integrity errors.	Fast search preserves usability during peak academic demand.
NFR-002	Nonfunctional	The system shall protect member and transaction data using authenticated role-based access, TLS in transit, and audit logging for privileged actions.	High	Unauthenticated requests are rejected; students cannot invoke librarian actions; all librarian mutations create timestamped audit entries; TLS 1.2+ is enforced in deployment tests.	Library records contain personal and financial information requiring confidentiality and accountability.